

Effect Sizes with 95% Confidence Intervals

Cohen's *d* (root-mean-square SD denominator) for between-group (Estonian vs. Russian) and within-group (2020 → 2023) comparisons across all 11 variables. Variables marked (inv.) have been scale-inverted so that positive *d* indicates more of the construct.

Table 1. Between-Group Comparisons (Estonian vs. Russian)

Variable	Year	Estonian M (SD), N	Russian M (SD), N	Cohen's <i>d</i>	95% CI	<i>p</i>	Sig
Superordinate Identity (inv.)	2020	3.55 (0.51), n = 701	2.92 (0.66), n = 602	+1.08	[+0.96, +1.20]	<.0001	***
Superordinate Identity (inv.)	2023	3.50 (0.60), n = 865	2.95 (0.79), n = 517	+0.79	[+0.68, +0.90]	<.0001	***
SD: Primary Out-group	2020	2.51 (0.88), n = 665	1.90 (0.80), n = 587	+0.74	[+0.62, +0.85]	<.0001	***
SD: Primary Out-group	2023	2.91 (1.04), n = 810	1.83 (0.77), n = 477	+1.18	[+1.06, +1.30]	<.0001	***
SD: General Out-group	2020	2.91 (1.01), n = 679	3.12 (1.02), n = 578	-0.21	[-0.32, -0.10]	0.0003	***
SD: General Out-group	2023	2.70 (0.90), n = 855	2.60 (0.81), n = 516	+0.11	[-0.00, +0.22]	0.049	*
Comparative Opportunity Assessment (inv.)	2020	3.44 (0.49), n = 679	3.86 (0.56), n = 603	-0.79	[-0.91, -0.68]	<.0001	***
Comparative Opportunity Assessment (inv.)	2023	3.29 (0.52), n = 844	3.68 (0.53), n = 518	-0.74	[-0.85, -0.62]	<.0001	***
Belief in Inevitable Conflict	2020	2.78 (0.57), n = 697	3.03 (0.58), n = 601	-0.44	[-0.55, -0.33]	<.0001	***
Belief in Inevitable Conflict	2023	2.72 (0.67), n = 855	3.21 (0.58), n = 507	-0.78	[-0.90, -0.67]	<.0001	***
Minority Support Inclusion (inv.)	2020	2.79 (0.70), n = 678	3.43 (0.59), n = 583	-1.00	[-1.12, -0.89]	<.0001	***
Minority Support Inclusion (inv.)	2023	2.67 (0.77), n = 844	3.46 (0.60), n = 505	-1.14	[-1.26, -1.02]	<.0001	***
Contact: Estonian Speakers (inv.)	2020	4.15 (0.98), n = 702	2.14 (1.14), n = 606	+1.89	[+1.76, +2.02]	<.0001	***
Contact: Estonian Speakers (inv.)	2023	4.26 (0.86), n = 865	2.71 (1.10), n = 521	+1.57	[+1.45, +1.69]	<.0001	***
Contact: Russian Speakers (inv.)	2020	2.02 (1.06), n = 701	4.21 (0.88), n = 607	-2.25	[-2.39, -2.11]	<.0001	***
Contact: Russian Speakers	2023	2.13 (1.10), n = 864	4.29 (0.80), n = 520	-2.25	[-2.38, -2.11]	<.0001	***

(inv.)							
Group ID Patterns	2020	2.95 (0.94), n = 630	2.69 (0.81), n = 591	+0.30	[+0.18, +0.41]	<.0001	***
Group ID Patterns	2023	2.94 (0.95), n = 787	2.81 (0.84), n = 498	+0.15	[+0.03, +0.26]	0.009	**
Territorial Attachment (inv.)	2020	3.76 (0.51), n = 697	3.57 (0.59), n = 593	+0.34	[+0.23, +0.45]	<.0001	***
Territorial Attachment (inv.)	2023	3.77 (0.51), n = 847	3.73 (0.53), n = 513	+0.09	[-0.02, +0.20]	0.116	ns
Contact: Out-group (inv.)	2020	2.02 (1.06), n = 701	2.14 (1.14), n = 606	-0.11	[-0.22, +0.00]	0.054	ns
Contact: Out-group (inv.)	2023	2.13 (1.10), n = 864	2.71 (1.10), n = 521	-0.53	[-0.64, -0.42]	<.0001	***

Table 2. Within-Group Change, 2020 → 2023

Variable	Group	2020 M (SD), N	2023 M (SD), N	Cohen's d	95% CI	p	Sig
Superordinate Identity (inv.)	Estonian	3.55 (0.51), n = 701	3.50 (0.60), n = 865	-0.10	[-0.19, +0.00]	0.059	ns
Superordinate Identity (inv.)	Russian	2.92 (0.66), n = 602	2.95 (0.79), n = 517	+0.04	[-0.08, +0.16]	0.524	ns
SD: Primary Out-group	Estonian	2.51 (0.88), n = 665	2.91 (1.04), n = 810	+0.42	[+0.31, +0.52]	<.0001	***
SD: Primary Out-group	Russian	1.90 (0.80), n = 587	1.83 (0.77), n = 477	-0.08	[-0.20, +0.04]	0.205	ns
SD: General Out-group	Estonian	2.91 (1.01), n = 679	2.70 (0.90), n = 855	-0.23	[-0.33, -0.12]	<.0001	***
SD: General Out-group	Russian	3.12 (1.02), n = 578	2.60 (0.81), n = 516	-0.56	[-0.68, -0.44]	<.0001	***
Comparative Opportunity Assessment (inv.)	Estonian	3.44 (0.49), n = 679	3.29 (0.52), n = 844	-0.30	[-0.40, -0.20]	<.0001	***
Comparative Opportunity Assessment (inv.)	Russian	3.86 (0.56), n = 603	3.68 (0.53), n = 518	-0.34	[-0.46, -0.22]	<.0001	***
Belief in Inevitable Conflict	Estonian	2.78 (0.57), n = 697	2.72 (0.67), n = 855	-0.09	[-0.19, +0.01]	0.077	ns
Belief in Inevitable Conflict	Russian	3.03 (0.58), n = 601	3.21 (0.58), n = 507	+0.32	[+0.20, +0.44]	<.0001	***
Minority Support Inclusion (inv.)	Estonian	2.79 (0.70), n = 678	2.67 (0.77), n = 844	-0.15	[-0.25, -0.05]	0.003	**
Minority Support Inclusion (inv.)	Russian	3.43 (0.59), n = 583	3.46 (0.60), n = 505	+0.04	[-0.08, +0.16]	0.490	ns
Contact: Estonian Speakers (inv.)	Estonian	4.15 (0.98), n = 702	4.26 (0.86), n = 865	+0.12	[+0.02, +0.22]	0.016	*
Contact: Estonian	Russian	2.14 (1.14), n = 606	2.71 (1.10), n = 521	+0.51	[+0.39, +0.63]	<.0001	***

Speakers (inv.)							
Contact: Russian Speakers (inv.)	Estonian	2.02 (1.06), n = 701	2.13 (1.10), n = 864	+0.10	[+0.00, +0.20]	0.042	*
Contact: Russian Speakers (inv.)	Russian	4.21 (0.88), n = 607	4.29 (0.80), n = 520	+0.09	[-0.03, +0.21]	0.133	ns
Group ID Patterns	Estonian	2.95 (0.94), n = 630	2.94 (0.95), n = 787	-0.01	[-0.12, +0.09]	0.816	ns
Group ID Patterns	Russian	2.69 (0.81), n = 591	2.81 (0.84), n = 498	+0.14	[+0.02, +0.26]	0.022	*
Territorial Attachment (inv.)	Estonian	3.76 (0.51), n = 697	3.77 (0.51), n = 847	+0.02	[-0.08, +0.12]	0.660	ns
Territorial Attachment (inv.)	Russian	3.57 (0.59), n = 593	3.73 (0.53), n = 513	+0.27	[+0.15, +0.39]	<.0001	***
Contact: Out-group (inv.)	Estonian	2.02 (1.06), n = 701	2.13 (1.10), n = 864	+0.10	[+0.00, +0.20]	0.042	*
Contact: Out-group (inv.)	Russian	2.14 (1.14), n = 606	2.71 (1.10), n = 521	+0.51	[+0.39, +0.63]	<.0001	***

Notes. Cohen's d uses the root-mean-square SD denominator: $d = (M_1 - M_2) / \sqrt{[(SD_1^2 + SD_2^2) / 2]}$. 95% CIs use the asymptotic standard error: $SE(d) = \sqrt{[(n_1 + n_2) / (n_1 n_2) + d^2 / (2(n_1 + n_2 - 2))]}$. All comparisons are independent-samples Welch's t -tests; the EIM is cross-sectional at each wave (different respondents in 2020 and 2023). Significance: *** $p < .001$, ** $p < .01$, * $p < .05$, ns = not significant. Variables marked (inv.) were scale-inverted (raw scale: lower = more of the construct) so that positive d in this table corresponds to more of the construct (e.g., more belonging, more contact). SD: General Out-group uses different items in 2020 (3 items, new immigrants) vs. 2023 (6 items, other Europeans + non-Europeans); within-group change estimates for that variable should be interpreted with caution.